

## Chapter

### Arithmetic Progression

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1. Which of the following is/are an Arithmetic Progression (A.P.)?

1. 1, 4, 9, 16,.....

2.  $\sqrt{3}, 2\sqrt{3}, 3\sqrt{3}, 4\sqrt{3}, \dots$

3. 8, 6, 4, 5

(a) only 1

(b) only 2

(c) only 2 and 3

(d) all 1, 2 and 3

SQP 2024

2. The 7<sup>th</sup> term of the given Arithmetic Progression (A.P.) :

$\frac{1}{a}, \left(\frac{1}{a} + 1\right), \left(\frac{1}{a} + 2\right) \dots$  is :

(a)  $\left(\frac{1}{a} + 6\right)$

(b)  $\left(\frac{1}{a} + 7\right)$

(c)  $\left(\frac{1}{a} + 8\right)$

(d)  $\left(\frac{1}{a} + 7^7\right)$

Main 2024

3. The  $n^{\text{th}}$  term of an Arithmetic Progression (A.P.) is  $2n + 5$ . The 10<sup>th</sup> term is:

(a) 7

(b) 15

(c) 25

(d) 45

Main 2023

4. The 5<sup>th</sup> term and the 9<sup>th</sup> term of an Arithmetic Progression are 4 and -12 respectively. Find:

- (a) the first term
- (b) common difference
- (c) sum of 16 terms of the *AP*.

Main 2023

5. Which term of the Arithmetic Progression (A.P.) 15, 30, 45, 60... is 300?  
Hence find the sum of all the terms of the Arithmetic Progression (A.P.)

Main 2023

6. The first term of AP is  $p$  and the common difference is  $q$ , then its 10th term is

- (a)  $q + 9p$
- (b)  $p - 9q$
- (c)  $p + 9q$
- (d)  $2p + 9q$

Comp 2023

7. Find the 20<sup>th</sup> term of an AP whose 3<sup>rd</sup> term is 7 and the seventh term exceeds three times the 3<sup>rd</sup> term by 2. Also find its  $n^{\text{th}}$  term ( $a_n$ ).

Comp 2023

8. The sum of the first 7 terms of an AP is 63 and that of its next 7 terms is 161. Find the AP.

Comp 2023

9. In an Arithmetic Progression (A.P.), if first term is 5, common difference is -3 and the  $n^{\text{th}}$  term is -7, then  $n$  is :

- (a) 5
- (b) 17
- (c) -13
- (d) 7

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**10. 70, 75, 80, 85 are the first four terms of an Arithmetic Progression, then the 10<sup>th</sup> term is :**

- |         |         |
|---------|---------|
| (a) 35  | (b) 25  |
| (c) 115 | (d) 105 |

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**11. The first four terms of an Arithmetic Progression (A.P.) whose first term is 4 and common difference is -6, are :**

- |                      |                   |
|----------------------|-------------------|
| (a) 4, -10, -16, -22 | (b) 4, 10, 16, 22 |
| (c) 4, -2, -8, -14   | (d) 4, 2, 8, 14   |

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**12. The first three terms of an Arithmetic Progression (A.P.) are 1, 9, 17 then the next two terms are :**

- |               |                   |
|---------------|-------------------|
| (a) 25 and 35 | (b) 27 and 37     |
| (c) 25 and 33 | (d) none of these |

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**13. If 73 is the  $n^{\text{th}}$  term of the arithmetic progression 3, 8, 13, 18 ..... then  $n$  is :**

- |        |        |
|--------|--------|
| (a) 13 | (b) 14 |
| (c) 15 | (d) 16 |

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## Solution

1.

**Ans: (b)** only 2

**Explanation:**

An arithmetic progression is a sequence in which the difference between consecutive terms is constant.

1. In 1, 4, 9, 16, ..., the differences are not equal. Hence, it is **not** an A.P.

2. In  $\sqrt{3}$ ,  $2\sqrt{3}$ ,  $3\sqrt{3}$ ,  $4\sqrt{3}$ , ..., the common difference is  $\sqrt{3}$ . Hence, it **is** an A.P.

3. In 8, 6, 4, 5, the differences are not constant. Hence, it is **not** an A.P.

**Thus, option (b) is the correct answer.**

2.

**Ans: (a)**  $\left(\frac{1}{a} + 6\right)$

**Explanation:**

From the given A.P.,

First term,

$$a = \frac{1}{a}$$

Common difference,

$$d = 1$$

The  $n$ th term of an A.P. is given by:

$$a_n = a + (n - 1)d$$

So,

$$a_7 = a + 6d$$

$$a_7 = \frac{1}{a} + 6$$

Thus, option (a) is the correct answer.

3.

Ans: (c) 25

Explanation:

Given the  $n^{\text{th}}$  meaning term of the A.P. is

$$a_n = 2n + 5$$

For the 10th term,

$$a_{10} = 2(10) + 5$$

$$a_{10} = 20 + 5$$

$$a_{10} = 25$$

Thus, option (c) is the correct answer.

4.

Ans:

We have

$$a_5 = 4 \quad \text{and} \quad a_9 = -12$$

Using the formula for the  $n^{\text{th}}$  term of an A.P.,

$$a_n = a + (n-1)d$$

So,

$$a + 4d = 4 \quad \dots\dots (1)$$

$$a + 8d = -12 \quad \dots\dots (2)$$

Subtracting (1) from (2),

$$4d = -16 \Rightarrow d = -4$$

Substituting  $d = -4$  in (1),

$$a - 16 = 4 \Rightarrow a = 20$$

(a) First term of the A.P. is **20**

(b) Common difference of the A.P. is **-4**

(c) **Sum of first 16 terms**

We know,

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$S_{16} = \frac{16}{2} [2(20) + 15(-4)]$$

$$S_{16} = 8(40 - 60)$$

$$S_{16} = 8(-20) = -160$$

**Final Answers:**

- First term = 20
- Common difference = -4
- Sum of first 16 terms = -160

**5.**

**Ans:**

Given A.P.: 15, 30, 45, 60, ...

First term,

$$a = 15$$

Common difference,

$$d = 30 - 15 = 15$$

Let 300 be the  $n^{\text{th}}$  term of the A.P.

Using the formula,

$$a_n = a + (n-1)d$$

$$15 + (n-1)15 = 300$$

$$15n = 300$$

$$n = 20$$

Hence, **300 is the 20th term** of the A.P.

**Sum of all the terms of the A.P.**

We know,

$$S_n = \frac{n}{2}(a + a_n)$$

$$S_{20} = \frac{20}{2}(15 + 300)$$

$$= 10 \times 315$$

$$= 3150$$

**Final Answers:**

- 300 is the 20th term of the A.P.
- Sum of all the terms of the A.P. is 3150

**6.**

**Ans: (c)  $p + 9q$**

**Explanation:**

We have

$$a = p \quad \text{and} \quad d = q$$

The  $n^{\text{th}}$  term of an A.P. is given by:

$$a_n = a + (n - 1)d$$

So, the 10th term is:

$$a_{10} = p + (10 - 1)q$$

$$a_{10} = p + 9q$$

**Thus, option (c) is the correct answer.**

7.

**Ans:**

Let the first term be  $a$  and the common difference be  $d$ .

Given,

$$a_3 = a + 2d = 7 \quad \dots\dots (1)$$

The 7th term exceeds three times the 3rd term by 2, so

$$a_7 = 3a_3 + 2$$

$$a + 6d = 3(7) + 2 = 23 \quad \dots\dots (2)$$

Subtracting (1) from (2),

$$4d = 16 \Rightarrow d = 4$$

Substituting  $d = 4$  in (1),

$$a + 8 = 7 \Rightarrow a = -1$$

20th term:

$$a_{20} = a + 19d$$

$$a_{20} = -1 + 19 \times 4 = 75$$

$n^{\text{th}}$  term:

$$a_n = a + (n - 1)d$$

$$a_n = -1 + 4(n - 1)$$

$$a_n = 4n - 5$$

**Final Answers:**

- 20th term = 75
- $n^{\text{th}}$  term =  $4n - 5$



8.

**Ans:**

Given,

$$S_7 = 63$$

We know,

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$63 = \frac{7}{2}[2a + 6d]$$

$$18 = 2a + 6d$$

$$a + 3d = 9 \quad \dots\dots (1)$$

Now, sum of the next 7 terms is

$$S_{8-14} = 161$$

$$S_{8-14} = \frac{7}{2}(a_8 + a_{14})$$

$$a_8 = a + 7d, \quad a_{14} = a + 13d$$

$$161 = \frac{7}{2}(2a + 20d)$$

$$23 = a + 10d \quad \dots\dots (2)$$

Subtracting equation (1) from equation (2),

$$14 = 7d \Rightarrow d = 2$$

Substituting  $d = 2$  in (1),

$$a + 6 = 9 \Rightarrow a = 3$$

Hence, the A.P. is:

|                     |
|---------------------|
| $3, 5, 7, 9, \dots$ |
|---------------------|

9.

Ans: (a) 5

Explanation:

We have

$$a = 5, \quad d = -3, \quad a_n = -7$$

Using the formula for the  $n^{\text{th}}$  term of an A.P.,

$$a_n = a + (n - 1)d$$

$$-7 = 5 + (n - 1)(-3)$$

$$-7 = 5 - 3n + 3$$

$$-7 = 8 - 3n$$

$$3n = 15$$

$$n = 5$$

Thus, option (a) is the correct answer.

10.

Ans: (c) 115

Explanation:

The given A.P. is:

$$70, 75, 80, 85, \dots$$

First term,

$$a = 70$$

Common difference,

$$d = 75 - 70 = 5$$

Using the formula for the  $n^{\text{th}}$  term of an A.P.,

$$a_n = a + (n - 1)d$$

For the 10th term,

$$a_{10} = 70 + (10 - 1) \times 5$$

$$a_{10} = 70 + 45$$

$$a_{10} = 115$$

Thus, option (c) is the correct answer.

11.

Ans: (c) 4, -2, -8, -14

**Explanation:**

We have

$$a = 4 \quad \text{and} \quad d = -6$$

First term:

$$a_1 = a = 4$$

Second term:

$$a_2 = a + d = 4 + (-6) = -2$$

Third term:

$$a_3 = a + 2d = 4 + 2(-6) = -8$$

Fourth term:

$$a_4 = a + 3d = 4 + 3(-6) = -14$$

Hence, the first four terms are 4, -2, -8, -14.

Thus, option (c) is the correct answer.

12.

Ans: (c) 25 and 33

**Explanation:**

The given A.P. is:

$$1, 9, 17, \dots$$

First term,

$$a = 1$$

Common difference,

$$d = 9 - 1 = 8$$

Fourth term,

$$a_4 = a + 3d = 1 + 3 \times 8 = 25$$

Fifth term,

$$a_5 = a + 4d = 1 + 4 \times 8 = 33$$

Thus, the next two terms are 25 and 33.

Hence, option (c) is the correct answer.

13.

Ans: (c) 15

**Explanation:**

The given A.P. is:

$$3, 8, 13, 18, \dots$$

First term,

$$a = 3$$

Common difference,

$$d = 8 - 3 = 5$$

Given,

$$a_n = 73$$

Using the formula for the  $n^{\text{th}}$  term of an A.P.,

$$a_n = a + (n - 1)d$$

$$73 = 3 + (n - 1)5$$

$$73 = 3 + 5n - 5$$

$$73 = 5n - 2$$

$$5n = 75$$

$$n = 15$$

Thus, option (c) is the correct answer.